

A generalized description of the signal size in extensive air shower detectors and its applications



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ABSTRACT

The number as well as the energy and angular distributions of particles in extensive air showers (EAS) depend on the stage of the shower development and the distance to the shower axis. In this work we derive an analytic parameterization of the particle distributions at ground from air shower simulations convolved with the response of a surface detector array. Shower particles are classified into four components according to the shower component they belong to: the muonic component, the electromagnetic component stemming from muon interactions and muon decay, the purely electromagnetic component, and the newly introduced electromagnetic component from low-energy hadrons. Using this scheme, we will show that the total signal at ground level for different surface detectors can be described with minimal fluctuations with parameterizations depending on the primary energy, position of the shower maximum, and the overall number of muons in the shower. The simulation results for different combinations of primaries and hadronic interaction models are reproduced with an accuracy better than 5–10% in the range from 100 m to 2000 m from the shower core. This parameterization is then used as a Lateral Distribution ansatz to reconstruct showers in current EAS experiments. Since this ansatz depends on physical parameters, it opens the possibility to infer them from data.

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1. Introduction

The origin of ultra-high-energy cosmic rays (UHECR, with energies beyond 10^{18} eV) still remains a mystery. Experimental results [1–4] suggest that the UHECR flux is composed predominantly of hadronic primary particles. As charged particles, they suffer deflections in cosmic magnetic fields and do not point back directly to their sources. Instead, an indirect search for their origin is necessary: the precise measurement of the energy spectrum, an estimation of the mass composition and its evolution with energy, and anisotropies in the arrival directions are the three main handles on disentangling this century-old problem.

Due to the low flux at ultra-high energies, the detection of UHECRs can only be achieved by measuring extensive air showers (EAS), cascades of secondary particles resulting from the interaction of the primary cosmic rays with the Earth's atmosphere. The measurements of the cosmic ray energy and mass rely on a good understanding of this phenomenon.

The electromagnetic component of extensive air showers of very high energy is characterized by a number of important fea-

tures, see for example [5–8]: the energy spectrum and angular distribution of electrons and gamma rays depend mainly on the shower age and the distance to the shower axis regardless of the primary particle. To exploit these properties in the analysis of data of modern cosmic ray detectors, it is necessary to extend the concept to the muonic shower component. Using simulations it was found that the longitudinal profile of muons in an air shower has a common shape for a wide variety of hadronic models and primaries (significant differences were only found in the normalization and depth of the shower maximum). Based on this finding, a parameterization of the expected signal in surface detector stations, such as the ones employed at the Pierre Auger Observatory, had been developed for a distance of 1000 m from the shower core [9].

In this paper, we explain a new description of air showers. In contrast to the previous parameterization, we introduce a new shower component which accounts for sub-showers of large transverse momentum initiated by neutral pions produced in hadronic interactions at low energy. This new component exhibits a natural scaling with the muon production in the shower: the neutral pions are produced together with charged pions that decay into muons. We explicitly use this relationship.

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2. Air shower simulations

2.1. Detector simulations

We used Geant4 [10] to simulate three types of detectors to be exemplary for modern UHECR experiments: the Pierre Auger water-Cherenkov detectors (WCD) and scintillation detectors above as well as below ground.

The water-Cherenkov detectors are comprised of 27 strips of plastic scintillator (1 cm thick, 4 cm wide, and 180 cm long; total area: 1.94 m²) placed on top of the water-Cherenkov detectors. The strips are complemented with an optical fiber that collects and guides scintillation photons. The number of photoelectrons in the connected PMT depends on the energy deposit and the position of the hit. The detector is similar to the AMIGA setup [12], but placed above ground. This is a configuration similar to the envisaged upgrade of the Pierre Auger Observatory [13].

Full Geant4 simulations of the detector response are performed in a grid of incident particle type, energy, and direction. The resulting photoelectron distributions are tabulated. The response is then calculated through an interpolation of the distributions. The signal unit in these two types of detectors is the VEM (Vertical Equivalent Muon), the average signal released by vertical incident muons with an energy of 1 GeV.

In the case of the water-Cherenkov detectors, a zenith angle grid from 0° up to 88° was utilized. In the case of the scintillators, the zenith angle grid only reached 70°. We allow an extrapolation up to 80°. In both cases, the bin size of the grid is 2°.

The underground scintillators are 2.25 m below ground with a soil density of 2.4 g/cm³. They are plastic scintillators of the same type as the ones used at the surface. However, a larger area is considered, using 32 strips (4 cm wide and 400 cm long; total area: 5.12 m²), similar to the unitary detector used in AMIGA [12]. In this case, the unit signal is a muon hit. The muon hit probability is calculated using Geant4 in a grid of muon energy and direction, and interpolated to get a uniform response covering the whole parameter space. Electromagnetic particles are not simulated for this type of detector.

2.2. Shower simulations

The shower library used in this work was simulated with CORSIKA v6890 [14]. Proton and iron primaries are simulated with the high-energy hadronic interaction model QGSJetII-03 [15] as well as with EPOS1.99 [16]. The low-energy interaction model FLUKA [17] was used, with a transition energy at 100 GeV. The thinning level is 10⁻⁶ with weight limits of 10⁻¹⁵ E/eV for electromagnetic particles and 10⁻¹⁷ E/eV for hadrons. The ground level is set at 1452 m. The magnetic field at the location of the Pierre Auger Observatory is taken into account. Electrons/gammas, muons and hadrons are followed in the CORSIKA simulations down to energies of 250 keV, 50 MeV and 100 MeV, respectively.

120 showers per energy and zenith angle were simulated. Within each set, 10 showers were simulated per monthly atmospheric model. The following zenith angles are considered: 0°, 12°, 25°, 36°, 45°, 53° and 60°. The simulated energies are 10^{18.6} eV, 10¹⁹ eV, 10^{19.5} eV and 10²⁰ eV.

The signal in a detector is calculated using the resampling procedure described in [18]. To reduce statistical fluctuations, all CORSIKA entries in a sampling area have been used to calculate the signals.

The shower plane is the plane perpendicular to the shower axis that contains the point at ground for which the signal is being evaluated. The Ψ angle is the azimuthal angle of this point in the shower plane ($\Psi=0^\circ/180^\circ$ corresponds to upstream/downstream).

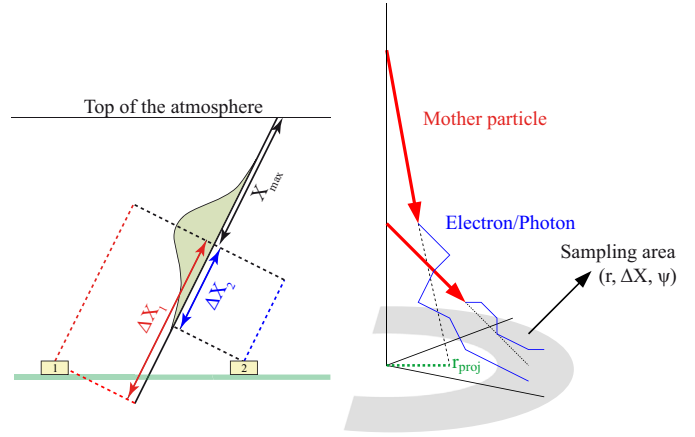


Fig. 1. Definition of the distance of a detector station to the shower maximum (left) and sketch to illustrate the definition of an electromagnetic particle from a jet (right).

We use a Cartesian coordinate system with the xy axes in the shower plane and the z axis pointing upward. The distance to the shower core r is calculated in the shower plane.

The sampling areas used to calculate the detector signals are slices of rings in the shower plane. They are centered at 100, 200, 400, 800, 1000, 1500, 2000 and 2500 m. The half width of each sampling area is 0.05 in log₁₀ scale. The binning in Ψ is: 0° $\triangleq$ [-30°, 30°); 45° $\triangleq$ [30°, 60°) and [300°, 330°); 90° $\triangleq$ [60°, 120°) and [240°, 330°); 135° $\triangleq$ [120°, 150°) and [210°, 240°); 180° $\triangleq$ [150°, 210°). We make use of the fact that the expected particle densities are symmetric to first order: $\rho(r, \Psi) = \rho(r, 360^\circ - \Psi)$.

The stage of shower development is quantified with the variable ΔX (see Fig. 1 (left) and [9]): the integral of the atmospheric density along the shower axis between the height at the projected detector position and the height of the electromagnetic shower maximum.

2.3. Shower components

Simulations show that a significant fraction of the particles arriving at a detector at a large distance from the shower core stems from hadronic interactions at low-energy, producing secondary particles at large angles with respect to the shower axis. Introducing the four shower components (a) the muonic component, (b) the electromagnetic component stemming from muon interactions and muon decay, (c) the purely electromagnetic component, and (d) the electromagnetic component from low-energy hadrons (jet component), we find a much more robust parameterization of showers compared to the findings in [9].

The newly introduced *jet* component (d) refers to electromagnetic particles for which the spatial distribution at ground is determined by the momentum of the mother particle at the last hadronic interaction as illustrated in Fig. 1 (right). The history version of CORSIKA v.6980 [19] has been used in the following. Each particle arriving at ground level is saved together with the information of the *mother* and *grandmother* particle at the last hadronic interaction.

The direction of the mother particle at the last interaction point is extrapolated to ground level where the distance to the shower core, r_{proj} , of the impact point is calculated. The left panel in Fig. 2 shows the distribution of r_{proj} for electromagnetic particles falling in sampling areas centered around a distance r_{dist} of 100, 200, 400 and 800 m, respectively. The results correspond to a proton shower, X_{max} is 1021 g/cm², the zenith angle 45° and the energy

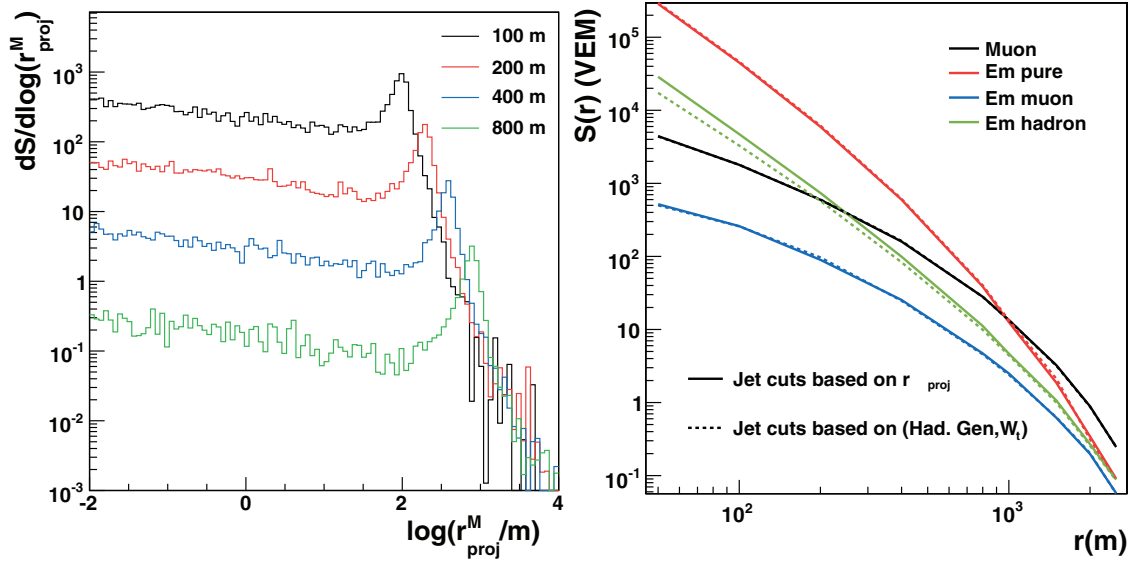


Fig. 2. Left panel: The distribution of r_{proj} for one proton shower and electromagnetic particles falling in sampling areas centered around a distance r_{dist} of 100, 200, 400 and 800 m, respectively. Right panel: The lateral distribution function for the 4 components for the same shower of the left panel.

10^{19} eV. A clear peak is seen in the distribution of r_{proj} close to a distance of r_{dist} . This peak can only be caused by sub-showers for which the lateral displacement of shower particles at ground level is dominated by the transverse momentum of the particle that initiated the sub-shower. The distributions for different core distances cut off at the core distance of the sampling area (the shower particles diffuse outwards). The ratio between the peak and the plateau of the distribution before the peak increases with core distance, showing the increasing importance of the *jet* component with increasing core distance. In the following, an electromagnetic particle is considered to originate from a jet if $\log_{10} r_{\text{proj}} > \log_{10} r_{\text{dist}} - 0.05$.

The shower library that will be used in this work has been simulated without the history flag of CORSIKA. Therefore, the jet component has to be inferred by indirect means. The particle weights in combination with the number of hadronic generations act as an effective way to discriminate the jet component (see Appendix A).

As an example, the signal of the four shower components is shown in the right panel of Fig. 2 for one shower simulated with the history option. The solid (dashed) lines correspond to the cuts based on r_{proj} and hadronic generation/weight, respectively. Differences at small core distances for the electromagnetic component from hadronic jets between the two cuts are evident. They are most likely due to background subtraction. The fall-off of the jet component reveals a different radial dependence with respect to the purely electromagnetic component.

3. Signal parameterization

In this section, we describe the procedure to derive the signal parameterization. It is carried out for the three detector types described before. For the sake of simplicity, when describing the model ingredients, only results for the water-Cherenkov detectors are shown. The model is tuned to proton QGSJetII-03 simulations (our reference) in the zenith range $0^\circ - 60^\circ$ ($0^\circ - 45^\circ$) for the water-Cherenkov detector (surface and underground scintillator).

An EAS will be characterized by its energy E , depth of maximum in shower development X_{max} , zenith angle θ , ground air density $\rho_{\text{ground}}^{\text{air}}$, and number of muons N_μ . The relative quantity N_μ attributed to a shower is defined as the ratio of the muonic signal to the predicted value for a proton QGSJetII-03 shower with the same

energy and X_{max} . It is evaluated at a radial distance of 1000m and azimuth of 90° , so there is a unique number for each shower.

The average signal released by a shower with a given set of $(E, X_{\text{max}}, N_\mu, \theta, \rho_{\text{ground}}^{\text{air}})$ in a ground detector at (r, Ψ) will be described by

$$\begin{aligned}
 S(r, \Psi | E, X_{\text{max}}, N_\mu, \theta, \rho_{\text{ground}}^{\text{air}}) &= \sum_{i=1, \dots, 4} S_0^i(r, \Delta X | E) \cdot f_{\text{mod}}^i(r, \Psi | \theta) \\
 &\cdot f_{\text{atm}}^i(r | \rho_{\text{ground}}^{\text{air}}) \cdot f_{\text{conv}}^i(r, \Delta X, \Psi | \theta) \\
 &\cdot f_{N_\mu, \text{fluct}}^i(r | N_\mu), \quad (1)
 \end{aligned}$$

where ΔX itself depends on $(r, \Psi, \theta, X_{\text{max}})$ and i refers to the four different shower components. The different terms of this equation are explained in the following:

- $S_0 \cdot f_{\text{mod}}$ is the signal deposited by a shower simulated without ground in an *ideal* detector: a detector with the same projected area A_v regardless of the incoming direction of shower particles and a response function as for vertical incident particles. $S_0 \cdot f_{\text{mod}}$ for showers simulated in a uniform atmosphere does not, by construction, have dependencies on (θ, Ψ) . In a *realistic* atmosphere, it has, for the reasons that will be explained below.
- $f_{\text{atm}}^i(r | \rho_{\text{ground}}^{\text{air}})$ is a factor that takes into account seasonal and day/night changes in the atmospheric profiles.
- f_{conv} is the conversion factor to a *realistic* detector. It takes into account changes in the detector response and projected area with the incoming direction of the shower particle, and the truncation of the distribution of incoming directions of shower particles due to the presence of the ground (no upward going particles).¹
- $f_{N_\mu, \text{fluct}}^i$ is a factor that takes into account the correlation between the component signal with N_μ .

For a given core distance and energy, Monte Carlo (MC) simulations have shown that the longitudinal development (i.e. its dependence on ΔX) of $S_0 \cdot f_{\text{mod}}$ can be well characterized by a Gaisser-Hillas function [20]. It was also found that the normalization and

¹ The truncation of the shower development due to the presence of the ground also contributes to the difference between an *ideal* and a *realistic* detector. Its contribution is small compared to the effects mentioned.

perhaps the shape parameters depend on θ and Ψ . There are two plausible reasons for this:

- For the electromagnetic components: the lateral spread in length units at a given ΔX depends on the atmospheric profile observed by the shower (Molière radius effects). Since the atmospheric profile changes with zenith angle, dependencies on θ are expected. Also, in particular for very inclined showers, shower particles propagating upstream/downstream traverse different atmospheric profiles. Thus, dependencies on Ψ are expected as well.
- For the muonic component: ΔX might not be the appropriate quantity to describe the attenuation of this component for two reasons: a) the relevant process is muon decay and the decay probability scales with traversed geometrical distance instead of integrated density; b) due to the quasi-rectilinear propagation of muons, the attenuation parameter should not be calculated parallel to the shower axis but along the line that joins the observation point and the average muon production height.

In our formulation, $S_0 \cdot f_{\text{mod}}$ factorizes into two terms:

- $S_0(\Delta X)$: the attenuation for $\theta = 0^\circ$ has by definition no Ψ dependency.
- $f_{\text{mod}}^i(\theta, \Psi)$: a change in the normalization of the attenuation curve that depends on (θ, Ψ) due to the effects mentioned above.

Both terms have an expected dependency on core distance, but the dependency on energy is carried only by S_0 (the effects described above should not depend on energy).

It is plausible that not only the normalization of the longitudinal development of $S_0 \cdot f_{\text{mod}}$ changes with θ and Ψ , but also the shape. We have ignored this possibility in our formulation for two reasons: a) a dedicated shower library with a uniform X_{max} distribution that allows full ΔX coverage would be required to parameterize the longitudinal development for each zenith angle; b) data indicates X_{max} values between 500–1200g/cm² and deviations in the shape in this short ΔX range are small.

The only term in Eq. (1) which depends on the hadronic model or the primary is S_0 .² This term is parameterized using the reference hadronic model and primary, proton QGSJetII-03.

The formulation described here is necessary to separate the different effects and to find an analytical description of the signals.

3.1. Parameterization of f_{conv}

We define the density of particles ρ as $\frac{dN}{dA}$ where dN is the number of particles through a differential surface dA perpendicular to the shower particle direction. We also define p_z as the cosine of the angle subtended by the particle direction with the shower axis. The signal in a realistic detector is given by

$$S_{\text{real}}(r, \Delta X, \Psi|\theta) = \int_{p_z^{\text{cut}}(\theta_p=\theta_p^{\text{cut}}|\theta, \Psi)}^1 \frac{d\rho}{dp_z}(p_z)(p_z|r, \Delta X) A(\theta_p) T(\theta_p|r, \Delta X, p_z) dp_z, \quad (2)$$

where

- θ_p is the zenith angle of the shower particle that can be calculated from $(p_z, X_{00C2}; \theta, \Psi)$ as detailed in Appendix B.
- p_z^{cut} is the value of p_z for which $\theta_p^{\text{cut}} = 90^\circ$ (i.e. no upward going particles due to the presence of the ground). In Fig. 3 the dependence of p_z^{cut} on the zenith angle for different Ψ angles is

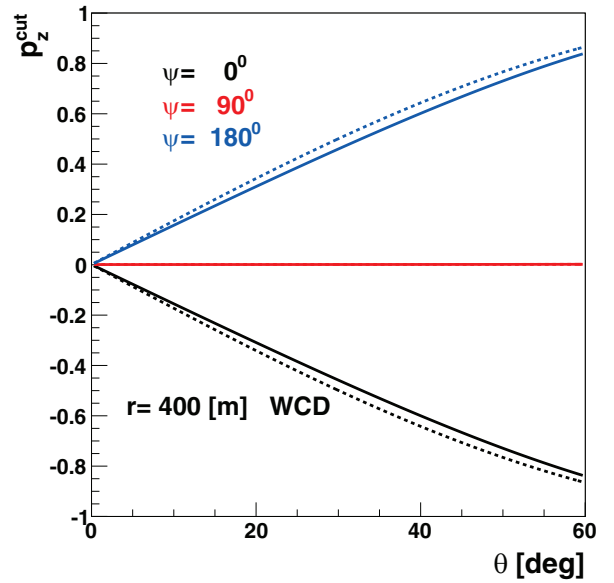


Fig. 3. The dependence of p_z^{cut} on the zenith angle for different Ψ angles.

shown. The solid lines correspond to the value adopted in this work. The dashed lines corresponds to $A_{\text{rad}}=1$ (see Appendix B).

- $\frac{d\rho}{dp_z}(p_z)$ is the p_z distribution whose shape and normalization depend primarily on $(r, \Delta X)$. Dependencies of the normalization on (θ, Ψ) exist, but are not indicated in the equation for simplicity.
- A is the projected area of the detector in the shower particle direction. It is given by

$$A(\theta_p) = A_v \cos \theta_p + A_h \sin \theta_p, \quad (3)$$

with A_h and A_v being the area of the detector in the horizontal and vertical direction.

- T is the convolution of the energy spectrum with the detector response, it is given by

$$T(\theta_p|r, \Delta X, p_z) = \int_{E_{\text{min}}}^{E_{\text{max}}} \frac{dN}{dE}(E|r, \Delta X, p_z) \text{VEM}(E, \theta_p) dE, \quad (4)$$

where $\text{VEM}(E, \theta_p)$ is the signal in VEM units of an incident particle with energy E and zenith angle θ_p . $\frac{dN}{dE}(r, \Delta X, p_z)$ is the normalized spectrum of particles that depend on $r, \Delta X$ and p_z .

The function $\frac{dS_0}{dp_z}$ is defined as

$$\frac{dS_0}{dp_z} = \frac{1}{N} \frac{d\rho}{dp_z}(p_z) A_v T_0, \quad (5)$$

where T_0 is $T(\theta_p = 0^\circ)$ and N is a normalization constant so the integral in the p_z range $(-1,1)$ is unity. N is just the total signal in an ideal detector given by

$$N = S_0(r, \Delta X) f_{\text{mod}}(r, \Psi|\theta), \quad (6)$$

with f_{mod} and S_0 as defined before.

Eq. (2) can then be rewritten as

$$S_{\text{real}}(r, \Delta X, \Psi|\theta) = N \int_{p_z^{\text{cut}}}^1 \frac{dS_0}{dp_z}(p_z|r, \Delta X) A_{\text{mod}}(\theta_p) T_{\text{mod}}(\theta_p|r, \Delta X, p_z) dp_z, \quad (7)$$

where A_{mod} is A/A_v and T_{mod} is T/T_0 .

In Appendix C and Appendix D an analytical parameterization of $\frac{dS_0}{dp_z}$ and $A_{\text{mod}} \cdot T_{\text{mod}}$ is presented.

² Hadronic model and primary dependencies of f_{mod} and f_{conv} have been checked and they are of second order.

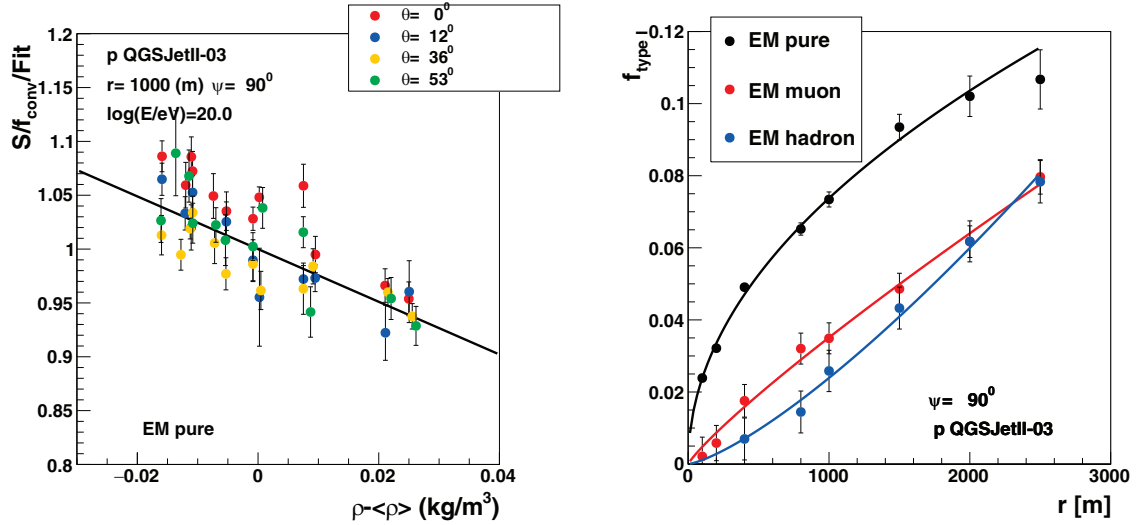


Fig. 4. Type I atmospheric effects, see text for more details.

The integral in Eq. (7) is equivalent to f_{conv} . It is evaluated in N steps. Instead of using bins in p_z , we use bins for which the integral of $\frac{dS_0}{dp_z}$ is $\frac{1}{N}$:

$$f_{\text{conv}} = \int_{p_z^{\text{cut}}}^1 \frac{dS_0}{dp_z} T_{\text{mod}}(\theta_p | r, p_z) A_{\text{mod}}(\theta_p) dp_z$$

$$= \sum_{i=0}^N \frac{T_{\text{mod}}(\theta_p | r, \langle p_z \rangle^i) A_{\text{mod}}(\theta_p)}{N}, \quad (8)$$

where $\langle p_z \rangle^i$ is the average p_z using the $\frac{dS_0}{dp_z}$ distribution in the corresponding bin. θ_p and $A_{\text{mod}} T_{\text{mod}}$ are evaluated at $\langle p_z \rangle^i$. For $N = 4$ we get an accuracy better than 5% for the parameter space of interest.

3.2. Atmospheric effects

Changes in the Molière radius due to changes in the atmospheric density a couple of radiation lengths above ground can modify the electromagnetic ground signal. For a given core distance, there are three possible effects:

- *Type I*: at a fixed $(\theta, \Psi, \Delta X)$ the density a couple of radiations lengths above ground fluctuates between day and night and between seasons.
- *Type II*: at a fixed $(\Psi, \Delta X)$ the density a couple of radiations lengths above ground changes with θ , leading to zenith angle dependent shifts.
- *Type III*: at a fixed $(\theta, \Delta X)$ shower particles arriving to sampling areas at different Ψ angles traverse different effective atmospheric profiles.

f_{atm} in Eq. (1) is only due to Type I effects, the other two effects are absorbed in the factor f_{mod} . In the following, we use the Monte Carlo simulations to estimate the Type I effects and use them to predict the Type II effects.

We fit $S(\Delta X)/f_{\text{conv}}$ to a Gaisser-Hillas function for each core distance, Ψ angle and energy. Zenith angles between 0° and 60° are used. The average ratio of the MC signal to the prediction is calculated in groups of showers that were simulated with the same atmospheric profile. The left panel in Fig. 4 shows these ratios as a function of over/under ground air density for four zenith angles, the pure electromagnetic component and for water-Cherenkov detectors. The solid line is a fit to

$$f_{\text{atm}} = 1 - f_{\text{type I}} \frac{\rho - \langle \rho \rangle}{0.03}, \quad (9)$$

where ρ is the density two radiation lengths above ground in kg/m^3 . For each core distance, we combine all energies and zenith angles to get a more precise estimation of the slope. Only $\Psi = 90^\circ$ is used, but we have not observed significant deviations between different Ψ angles. The slope is parameterized as a function of core distance, in the right panel of Fig. 4 the results for the three electromagnetic components are shown.

Using this parameterization it is possible to predict the Type II effects:

$$f_{\text{type II}}(r, \theta) = 1 - f_{\text{type I}}(r) \frac{\langle \rho(\theta) \rangle - \langle \rho(0^\circ) \rangle}{0.03} \quad (10)$$

The left panel in Fig. 5 shows $f_{\text{type II}}$ as a function of zenith angle for a core distance of 1000 m and the three electromagnetic components. Modulations with zenith angle at the level of 10% are therefore expected for the largest zenith angles considered.

The density two radiation lengths above ground is calculated using the zenith angle of the shower. This is the correct procedure for the pure electromagnetic component because of its diffusive characteristics. For the electromagnetic component stemming from muons and the electromagnetic component from the low-energy hadron component, and due to their jet-like behavior, such an approximation might not be correct. It is possible that the relevant density above ground for those components depends on (r, θ, Ψ) . This would contribute to the Type III atmospheric effects.

3.3. Parameterization of f_{mod}

We use the following ansatz for $f_{\text{mod}}(r, \Psi | \theta)$

$$f_{\text{mod}}(r, \Psi | \theta) = M_0(r, \theta) (1 + M_1(r, \theta) \cos \Psi), \quad (11)$$

where M_0 describes the change in normalization with (r, θ) for $\Psi = 90^\circ$, and M_1 is a simple modulation in $\cos \Psi$ with an amplitude that depends on zenith angle and core distance. M_0 is fixed to Eq. (10). M_1 is derived from a fit to MC simulations.

For each core distance, $S(\Delta X)/(f_{\text{conv}} \cdot f_{\text{mod}})$ is fitted to a Gaisser-Hillas function using all zenith angles. The free parameters are the Gaisser-Hillas parameters and a value of M_1 for each of the zenith angles included in the fit. The right panel in Fig. 5 shows an example of the inferred values: M_1 as a function of zenith angle for a core distance of 1000 m and all the shower components. These results are used to parameterize $M_1(r, \theta)$ using simple functional shapes (solid lines in the figure). Interestingly, M_1 for the muonic component can be as large as 10% for the largest zenith angle con-

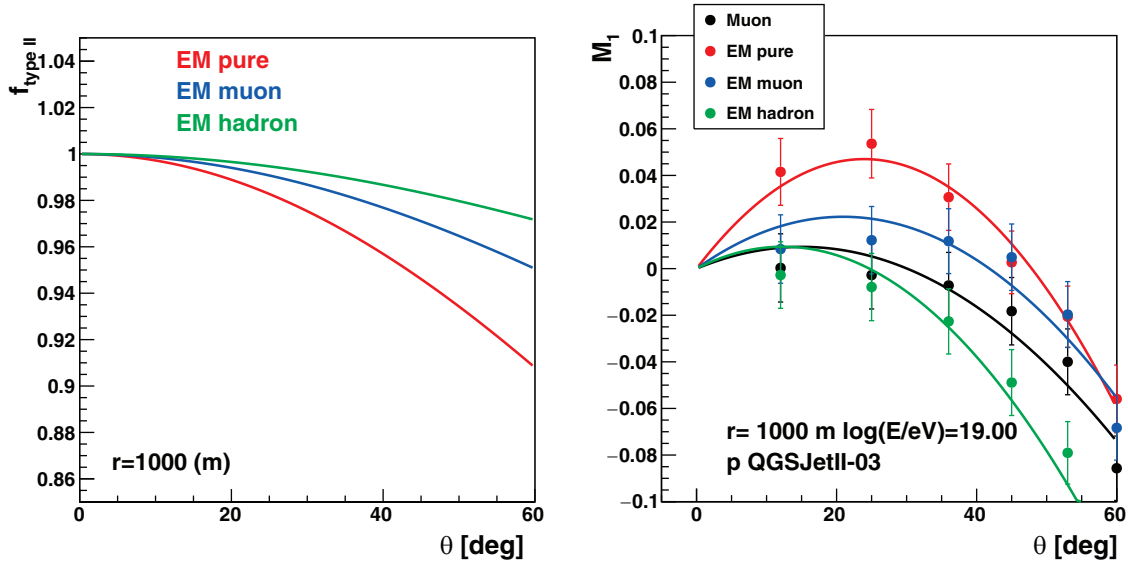


Fig. 5. Left panel: f_{typeII} as a function of zenith angle for a core distance of 1000m and the three electromagnetic components. Right panel: M_1 as a function of zenith angle for a core distance of 1000 m and all the shower components.

sidered. As mentioned before, this is due to ΔX not being the optimal parameter to describe the attenuation of the muonic component.

3.4. Parameterization of S_0^{ref}

S_0 , the signal in an *ideal* detector for $\theta = 0^\circ$, is estimated from the MC simulations for the reference primary and model (proton, QGSJetII-03). To increase the accuracy of the final parameterization we use the signals in a *realistic* detector divided by the parameterized values of f_{conv} and f_{mod} . The dependence on $(r, \Delta X, E)$ of this quantity is parameterized in the following.

For each core distance and shower component, S_0 is fitted to

$$S_0(\Delta X, E) = S_{\text{ref}} \left(\frac{E}{10^{19} \text{eV}} \right)^\gamma \left(\frac{\Delta X - \Delta X_0}{\Delta X_{\text{ref}} - \Delta X_0} \right)^{\frac{\Delta X_{\text{max}} - \Delta X_0}{\lambda(E)}} \cdot \exp \left(\frac{\Delta X_{\text{ref}} - \Delta X}{\lambda(E)} \right), \quad (12)$$

where $\lambda(E) = \lambda_0 + f_\lambda \log_{10}(E/10^{19} \text{eV})$. f_λ is fixed to 0 for all components except for the pure electromagnetic ($f_\lambda = 0.04$) and electromagnetic from the low-energy hadron component ($f_\lambda = -0.02$). The value of ΔX_0 is fixed to $-250, -500, 250$ and -800 g/cm^2 for the muonic, pure electromagnetic, electromagnetic from muon and electromagnetic from low-energy hadron components, respectively. S_{ref} is the value of the signal at ΔX_{ref} (400 g/cm^2 , close to the maximum for most of the components).

The solid lines in Fig. 6 show the result of the fit for a core distance of 1000 m and each of the shower components. Each point corresponds to a sampling area in a CORSIKA shower. Only sampling areas with $\Psi = 0^\circ, 90^\circ$ and 180° , and showers with energy of 10^{19} eV are shown. The different colors correspond to different zenith angles. The errors correspond to the Poisson fluctuations of the number of weighted particles used to calculate the signal. Since shower to shower fluctuations are larger than the error in the calculation of individual signals, we use a χ^2 -method in which the average ratio (MC/predicted) for sets of showers and sampling areas with the same (θ, E, Ψ) is used, allowing for 5% statistical fluctuations.

In Fig. 7, the dependence of S_{ref} , γ , ΔX_{max} and λ in Eq. (12) on core distance is displayed for each of the four shower components.

The lines correspond to our derived parameterization. The distinct behavior of the different components with $(r, \Delta X, E)$ becomes evident.

3.5. Parameterization of $f_{N_\mu \text{fluct}}$

In Fig. 8, the correlation between $S_{0,\text{em}}/S_{0,\text{em}}^{\text{ref}}$ and $S_{0,\mu}/S_{0,\mu}^{\text{ref}}$ is shown. $S_{0,\mu}^{\text{ref}}$ refers to the parameterized value of S_0 . The left, middle and right panels correspond to the pure electromagnetic component at $r = 100 \text{ m}$, and the electromagnetic from muon and from low-energy hadrons component at $r = 1000 \text{ m}$. θ is 36° , $\Psi = 90^\circ$ and the energy 10^{19} eV . The different colors indicate different combinations of primary and models. The solid line is a fit to the reference model and primary (proton, QGSJetII-03):

$$\frac{S_{0,\text{em}}}{S_{0,\text{em}}^{\text{ref}}} = 1 + \left(\frac{S_{0,\mu}}{S_{0,\mu}^{\text{ref}}} - 1 \right) \alpha. \quad (13)$$

This formula provides a good description of the reference, deviations for other combinations of models and primaries are regarded as intrinsic model systematics. The value of α does not show a dependence on core distance for the pure electromagnetic component and the electromagnetic component from muon decay. The derived values of α are -0.075 and 1 , respectively. In the case of the electromagnetic component from low energy hadrons, there is a mild dependence on core distance that is parameterized as

$$\alpha = A + C \exp \left(-B \frac{r}{1000 \text{m}} \right), \quad (14)$$

with $A = 1.25$, $B = 6$ and $C = -0.13$. $f_{N_\mu \text{fluct}}$ in Eq. (1) is then given by $f_{N_\mu \text{fluct}} = 1 + (N_\mu - 1) \alpha$ with the corresponding values of α . Note that N_μ is only exactly equal to $\frac{S_{0,\mu}}{S_{0,\mu}^{\text{ref}}}$ for a core distance of 1000 m and $\Psi = 90^\circ$.

4. The accuracy of the parameterization

The presented model was built using **only** MC simulations for the reference model and primary (proton QGSJetII-03). To establish the accuracy of the model in predicting the total signal for different combinations of primaries and models, we calculate the average ratio of the simulated signals to the prediction (Eq. (1)) using, for each shower, the corresponding values of energy, X_{max} and

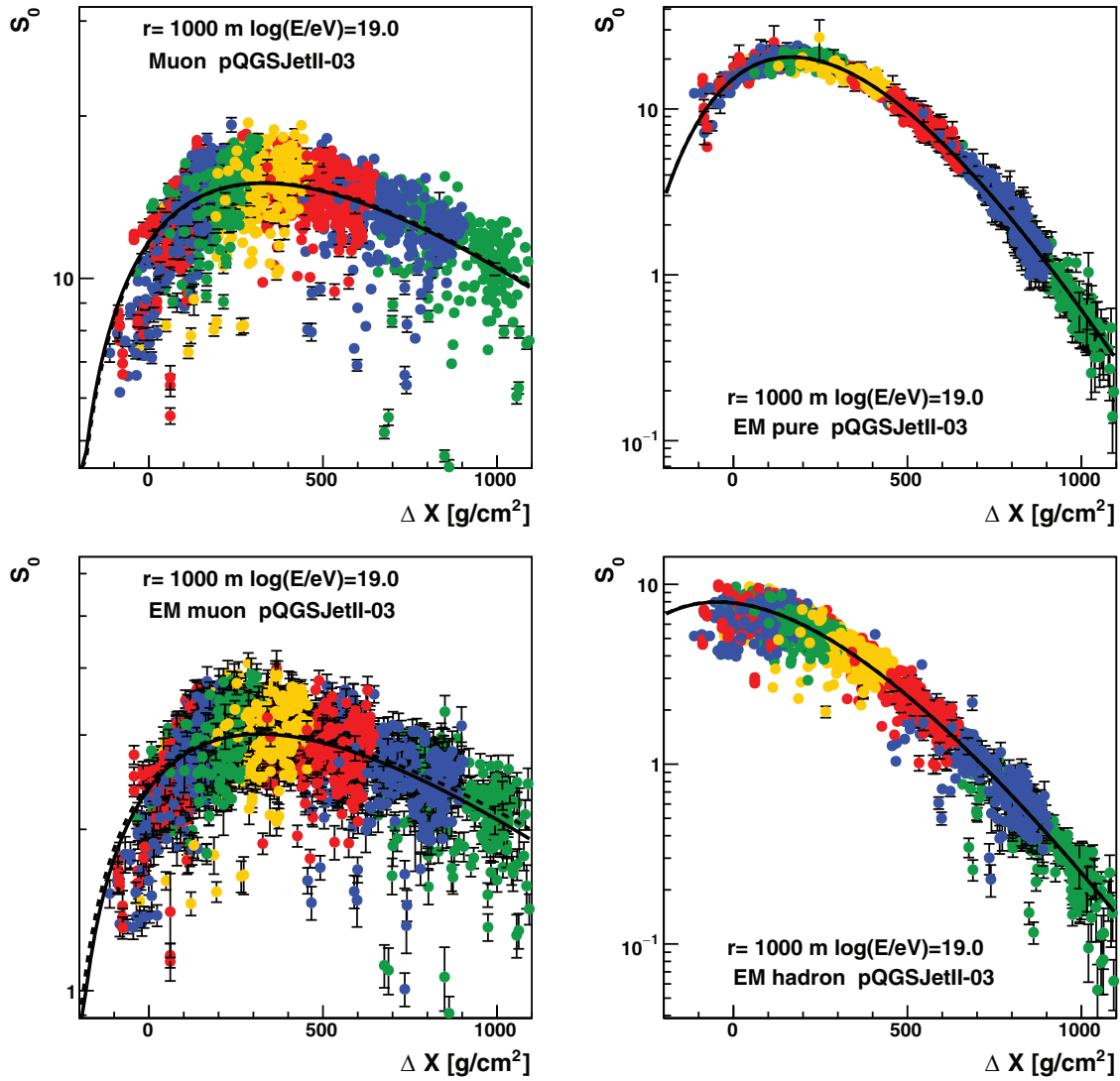


Fig. 6. $\frac{S}{S_0}$ as a function of ΔX for each of the shower components considered and a core distance of 1000 m. Each point corresponds to a sampling area in a CORSIKA shower. The different colors correspond to different zenith angles.

N_μ . Fig. 9 shows an example for $\theta = 25^\circ$ (45°), upper (lower) panels. Left, middle and right panels correspond to water-Cherenkov detectors, surface and underground scintillators. The Ψ angle is 90° . The average ratios are plotted as a function of core distance for two energies and different combinations of models and primaries. Similar results are obtained for all Ψ within the zenith angle range used to build the model: $[0^\circ, 60^\circ]$ ($[0^\circ, 45^\circ]$) for the water-Cherenkov detector (surface and underground scintillator). Deviations for the reference model and primary are due to inaccuracies of the parameterization to reproduce the MC results. These deviations have to be subtracted from the results for other primaries and models to estimate model/primary dependencies. The accuracy of the parameterization over the whole parameter space and all tested hadronic models and primaries is better than 10%.

In the case of underground scintillators, the total signal coincides with the muonic signal. There are deviations of $\pm 3\%$ for the reference model and primary (proton QGSJetII-03). On top of that, there are $\pm 5\%$ deviations due to intrinsic differences of the muon LDF between primaries and models, i.e. adopting a single value of N_μ for the shower is accurate within 5%.

5. Applications to EAS reconstruction

Current EAS experiments use two experimental techniques:

- Fluorescence detectors (FD): the longitudinal development of the electromagnetic component is measured by detecting the fluorescence light emitted by nitrogen molecules in the air that were excited by secondary electrons. The duty cycle is $\approx 10\%$, but this technique provides a calorimetric measurement of the primary energy together with a precise and unbiased measurement of the position of the shower maximum.
- Surface detectors (SD): the lateral profile of the shower front at ground level is measured. The SD has a duty cycle of almost 100% and is sensitive to the number of muons. The arrival time information of the shower particles can also be used to estimate the position of the shower maximum.

The experiments that are currently taking data (the Pierre Auger Observatory and the Telescope Array [21]) are hybrid experiments that use both techniques. The analysis of the SD data requires the calibration of its energy scale using FD measurements. The usual procedure in case of the Pierre Auger Observatory is the following:

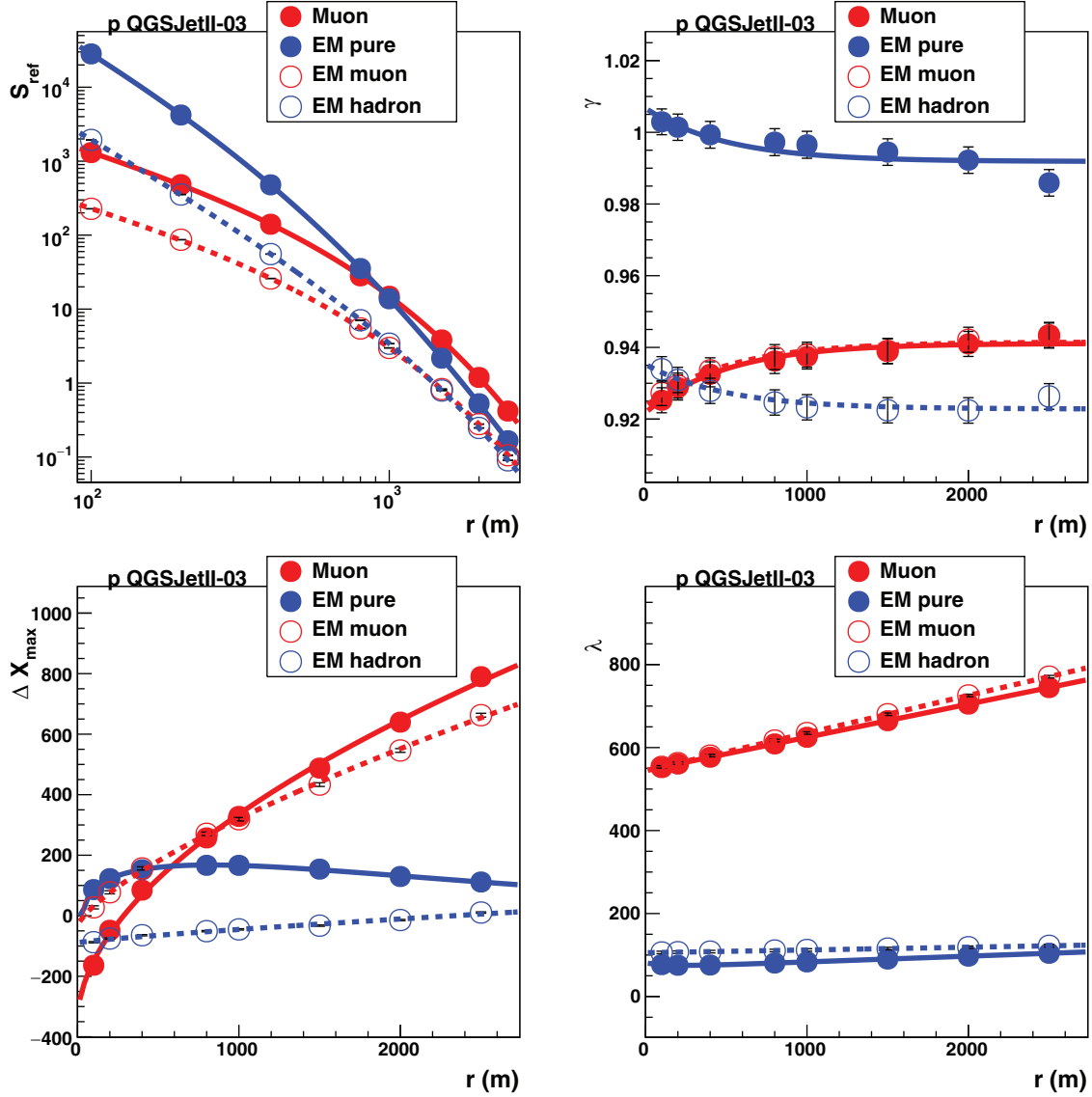


Fig. 7. The parameterization of S_{ref} , γ , ΔX_{max} and λ as a function of core distance for the four shower components, the reference model and primary (proton, QGSJetII-03), and the water-Cherenkov detector response.

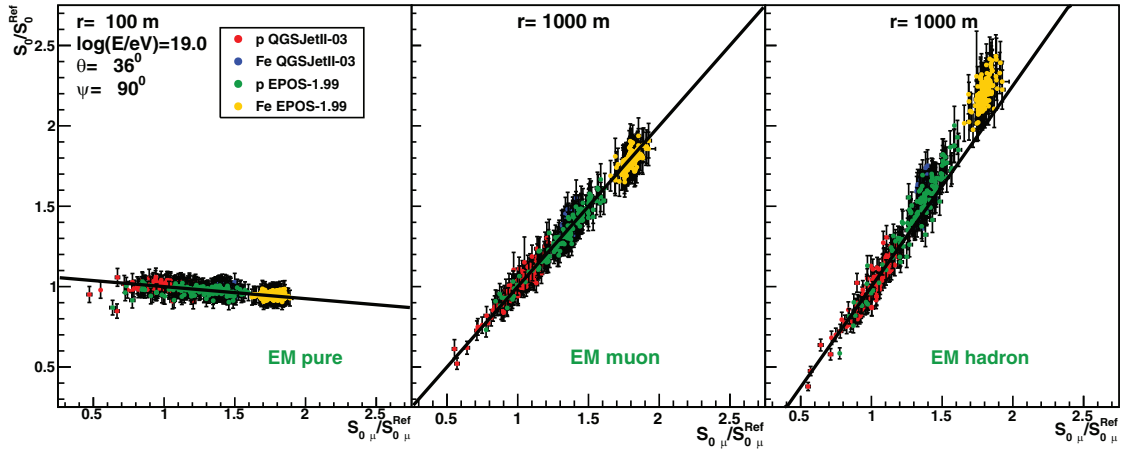


Fig. 8. The correlation between $S_{0,\text{em}}/S_{0,\text{em}}^{\text{ref}}$ and $S_{0,\mu}/S_{0,\mu}^{\text{ref}}$. The left, middle and right panels correspond to the pure electromagnetic component at $r=100\text{m}$, and the electromagnetic from muon and from low-energy hadrons component at $r=1000\text{m}$. θ is 36° , $\psi = 90^\circ$ and the energy 10^{19}eV . The different colors indicate different combinations of primaries and models.

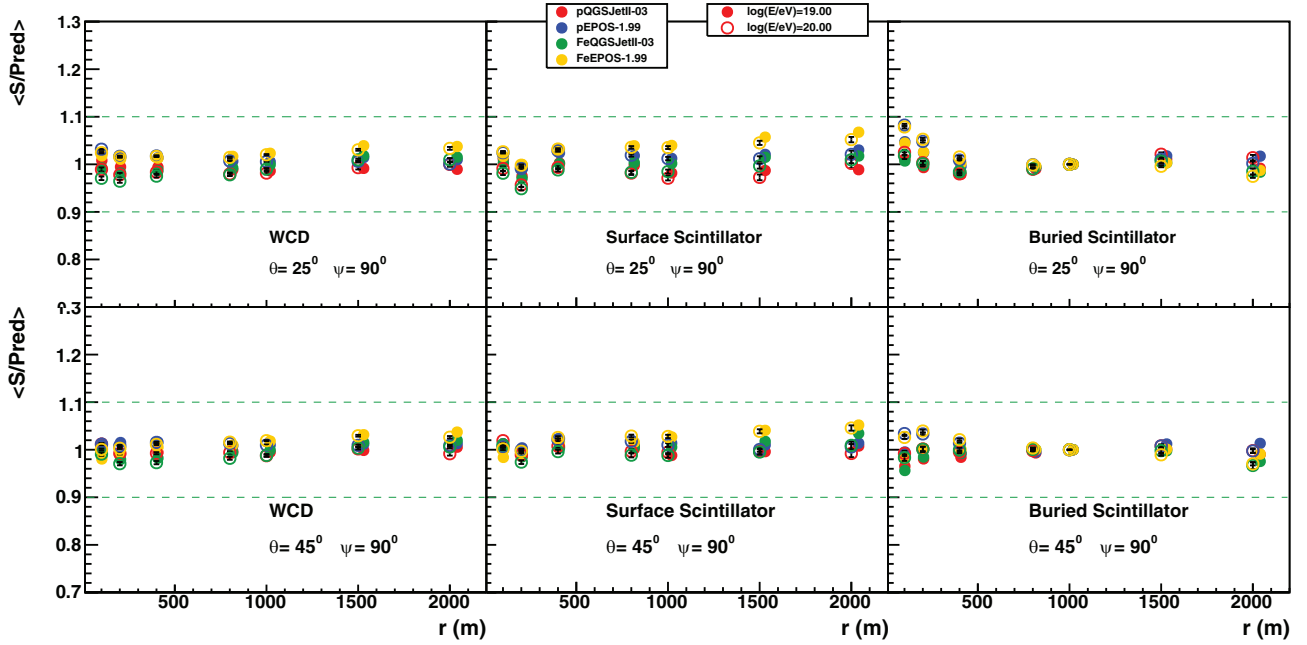


Fig. 9. The average ratio of the simulated signals to the prediction Eq. (1) as a function of core distance for $\theta = 36^\circ$. Left, middle and right panels correspond to water-Cherenkov detectors, surface and underground scintillators. The Ψ angle is 90° . Two energies and different combinations of models and primaries are shown.

- SD events are analyzed using an empirical model of the lateral distribution function (LDF). The shower size is the estimated signal at a given optimal distance $S(R_{\text{opt}})$, which depends on the spacing of detectors in the ground array.
- The dependence of $S(R_{\text{opt}})$ with zenith angle for a fixed energy is derived from SD data assuming that the primary cosmic ray flux does not have a zenith angle dependency. This yields for each event an estimator at a reference zenith angle: $S(R_{\text{opt}}|\theta_{\text{ref}})$.
- A subsample of SD events, which are also detected by FD (Hybrid events), is used to establish the relationship between $S(R_{\text{opt}}|\theta_{\text{ref}})$ and E_{FD} .

The signal parameterization presented in this work is *similar* to a Monte Carlo based LDF which depends on physical parameters and is valid for a wide range of hadronic models and primaries. It accounts properly for atmospheric and ground effects and the different atmospheric attenuation of each of the shower components. Shower to shower fluctuations at a fixed energy are accounted for by its dependence on X_{max} and N_μ . Moreover, it is analytical, which allows its usage in the minimization of Likelihood functions during shower reconstruction. As any other *empirical* LDF used in current SD experiments, it can be tested with real data by different means. In the following, we will present example applications.

Simulated events in triangular arrays with 750 m and 1500 m spacing (like at the Pierre Auger Observatory) were generated from CORSIKA showers, using the detector simulation described before, and the resampling procedure given in [18] to account for sampling (thinning) weights of the recorded ground particles. Water-Cherenkov and surface scintillators were simulated at each detector position. Fig. 10 shows an example of one event in the 750 m array.

Station trigger and saturation effects are taken into account³. The FD detector is not simulated. The FD energy, X_{max} and shower direction are taken from the *true* values fluctuated with Gaussian distributions of 10%, 20 g/cm² and 0.25° width.

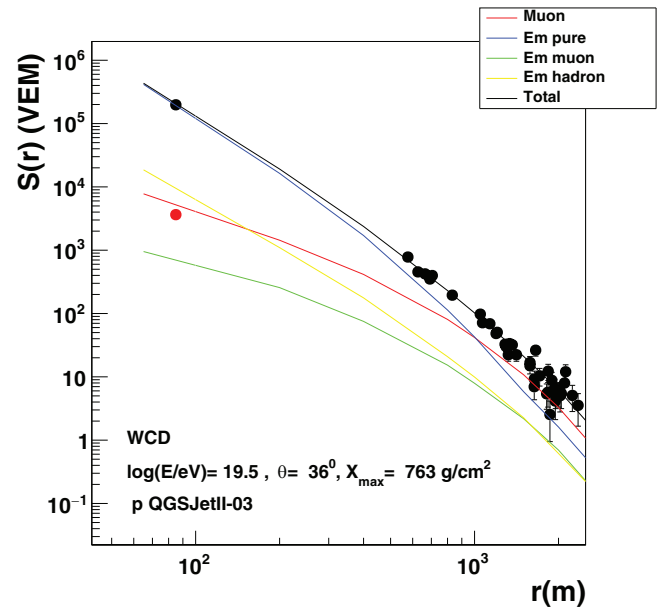


Fig. 10. An example of an event with a randomly selected core position in an array of water-Cherenkov detectors with 750 m spacing.

In Section 5.1 we show a procedure to derive N_μ on an event by event basis using Hybrid events in experiments with SD and FD detectors. Using $\langle N_\mu \rangle$ from the previous procedure together with the $\langle X_{\text{max}} \rangle$ measured by FD, we present in Section 5.2 a procedure to derive the energy for any SD event, independent of FD measurements. We use water-Cherenkov detectors for SD, but the procedures presented are analogous for an array of surface scintillators. Finally, in Section 5.3 we present a procedure to derive N_μ on an event-by-event basis in an SD array with two types of detectors at each array position: surface scintillators and water-Cherenkov detectors.

³ The saturation threshold is $\approx 6,000\text{VEM}$ (4,000VEM) for surface scintillators (water-Cherenkov detectors).

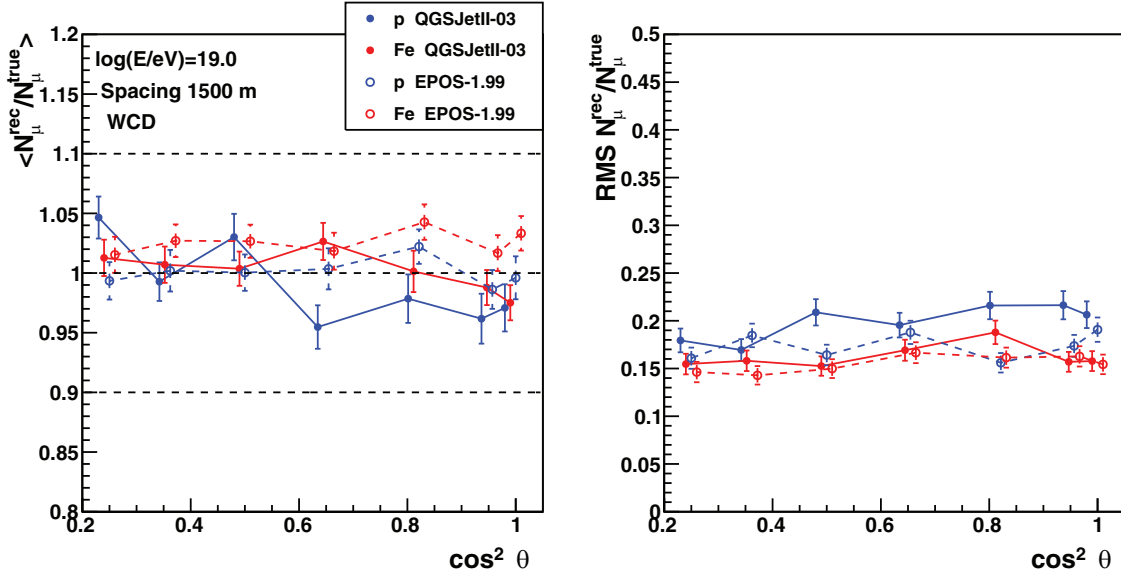


Fig. 11. The mean and RMS of $N_{\mu}^{\text{rec}}/N_{\mu}^{\text{true}}$ as a function of zenith angle at 10^{19} eV for the N_{μ} FD based reconstruction described in the text.

An advantage of the presented procedures is that results are given in terms of physical quantities. This allows the direct comparison between EAS experiments, and also cross calibrations.

5.1. Hybrid events: N_{μ}^{FD} and its correlation with X_{max}

Showers with zenith angles larger than 60° detected by FD and SD in the Pierre Auger Observatory have been used in the past to estimate the muon content in EAS [22]. At large zenith angles, the systematics in the modeling of the electromagnetic component are small due to its large absorption in the atmosphere. This estimation can be extended to vertical showers ($[0^\circ, 60^\circ]$) using the presented parameterization to subtract the electromagnetic component at ground, which is determined by X_{max} and energy with small deviations that can be parameterized using N_{μ} . This would enhance the available statistics by a factor of four.

The definition of N_{μ} adopted in this work agrees with the N_{19} parameter used in analyses of inclined showers at 10 EeV [22]. This allows a direct comparison between the reconstructions.

Hybrid MC events corresponding to an array of water-Cherenkov detectors with 750 and 1500 m spacing are reconstructed as follows. The signals are fitted to the signal parameterization using a likelihood method. The reconstruction is done in two steps with N_{μ} and core positions as free parameters. In the first step, all triggered stations with signals larger than 3VEM are used, the X_{max} and energy are fixed to the reconstructed FD values. In the second step, all detectors where the predicted signal is larger than 3VEM and the core distance smaller than 2500m are used, the X_{max} and energy are constrained to the reconstructed FD values using a Gaussian distribution with a width of 20 g/cm² and 10% respectively. The shower geometry, i.e. (θ, ϕ) is taken from the Hybrid reconstruction. The air density at ground for each event is used to correct for atmospheric effects.

Fig. 11 shows the mean and RMS of $N_{\mu}^{\text{rec}}/N_{\mu}^{\text{true}}$ as a function of zenith angle at 10^{19} eV. Results for different combinations of primaries/models and an array spacing of 1500m are shown. Biases due to assumptions on the primary and hadronic model are below 5%. The resolution is 15–20%.

In Fig. 12 the mean and RMS using all zenith angles is shown as a function of energy. The uncertainty corresponds to the maximum difference found with different combinations of primaries and models. Results for two array spacings are given. The recon-

struction of N_{μ} has similar performance than the inclined shower estimation in [22].

An advantage of working at small zenith angles is that the FD X_{max} estimation is more precise.⁴ This, together with the enhancement of the statistics, opens the possibility of studying for the first time the correlation between N_{μ} and X_{max} . Fig. 13 shows this correlation in reconstructed MC simulations data sets with 50% of proton and iron showers. The energy is $10^{19.5}$ eV and the array spacing 1500 m. The left and right panels correspond to the hadronic models QGSJetII-03 and EPOS1.99, respectively. Blue (red) markers correspond to proton (iron) primaries. Black markers correspond to the average in X_{max} bins. The black line corresponds to

$$N_{\mu} = \langle N_{\mu} \rangle \frac{1 + \frac{1}{2\pi} \arctan \left(\frac{X_{\text{max}}^{\text{pFe}} - X_{\text{max}}}{40 \text{ g/cm}^2} \right)}{1 + \frac{1}{2\pi} \arctan \left(\frac{X_{\text{max}}^{\text{pFe}} - \langle X_{\text{max}} \rangle}{40 \text{ g/cm}^2} \right)} \quad (15)$$

where $X_{\text{max}}^{\text{pFe}}$ is the average X_{max} of a mixture of 50% proton/iron. This function gives a good description of both models: the difference in $(\langle N_{\mu} \rangle, \langle X_{\text{max}} \rangle)$ between proton and iron primaries of about (40%, 80 g/cm²) is rather model independent. Intermediate primaries lie also along this line.

Different mixtures of p/He/N/Fe are also described by Eq. (15). $X_{\text{max}}^{\text{pFe}}$ is the only parameter needed to describe the correlation between N_{μ} and X_{max} . If the cosmic ray flux contains both proton and iron primaries, this parameter can be extracted from data.

5.2. SD events: energy reconstruction

In this section, we consider a cosmic ray flux with 50% proton and iron primaries each. Different mixtures have been found to yield similar results. Water-Cherenkov detectors with two array spacings are considered.

⁴ Small changes in the shower direction at large zenith angles translates into large changes in the atmospheric profile density, and, therefore induce errors in the X_{max} estimation.

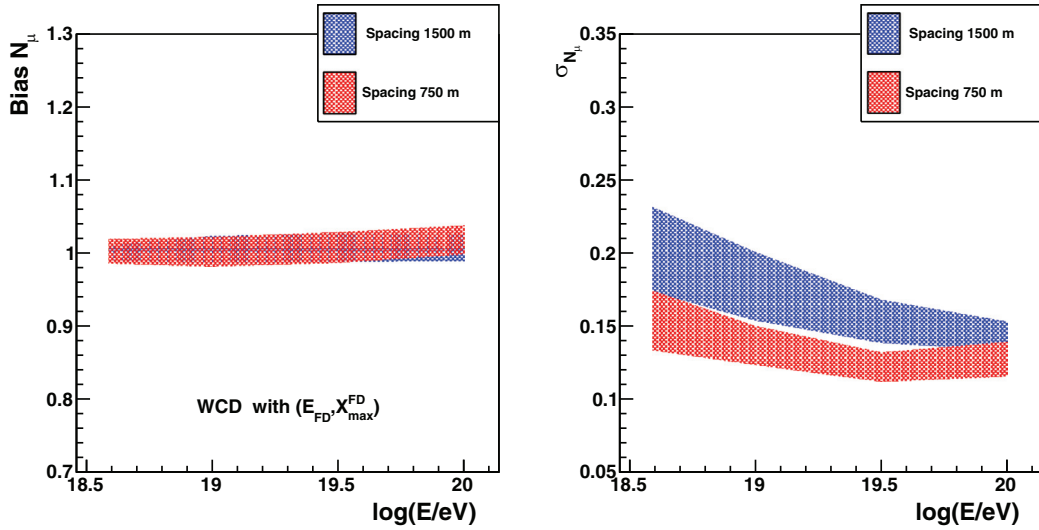


Fig. 12. The mean and RMS of $N_{\mu}^{\text{rec}}/N_{\mu}^{\text{true}}$ averaged over all zenith angles as a function of energy for the FD based N_{μ} reconstruction described in the text.

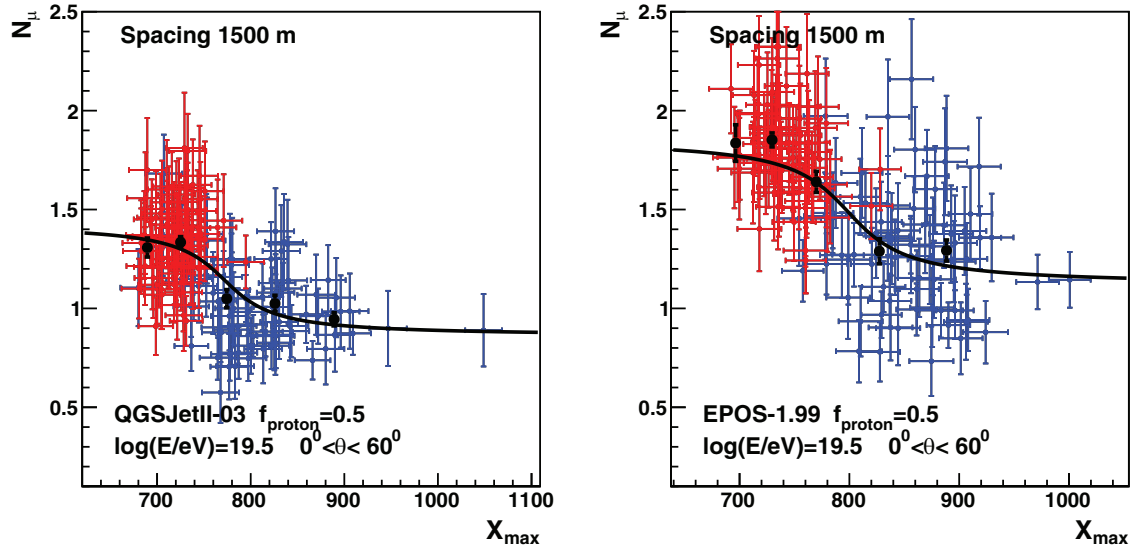


Fig. 13. The correlation between N_{μ} and X_{max} in reconstructed sets of shower simulations with 50% proton and iron showers each. Blue (red) markers correspond to proton (iron) primaries. Black markers correspond to the average in X_{max} bins. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The dependence of the reconstructed $\langle N_{\mu} \rangle$ on zenith angle and energy from Hybrid events is fitted to the following ansatz⁵:

$$\langle N_{\mu} \rangle(E, \theta) = N_{\mu,0} (1 + \alpha (\sec \theta - 2)) \left(\frac{E}{E_0} \right)^{\gamma}, \quad (16)$$

with α and γ as free parameters. The average X_{max} of the mixture is fitted with the following function:

$$\langle X_{\text{max}} \rangle(E) = X_{\text{max},0} + ER \log_{10} \left(\frac{E}{E_0} \right), \quad (17)$$

with the elongation rate ER. SD events are now reconstructed using the signal parameterization with $(N_{\mu}, X_{\text{max}})$ fixed to the values in Eqs. (16) and (17). The shower direction, energy and core position are free parameters.

⁵ The dependence on zenith angle has been introduced to account for possible differences between the *real* muon energy spectrum w.r.t. the reference model and primary. Different zenith angles imply different energy thresholds for the muons arriving at ground.

The direction of the shower is reconstructed in two steps using a χ^2 -minimization. In the first step, the start time of the stations in the triangle with the largest sum of signals is fitted to a plane front. In a second step, all stations with signal larger than 3VEM are used and curvature and altitude corrections are applied. The time variance model is adopted from [23].

The energy and core position is reconstructed using a likelihood minimization in two steps as described in the previous section.

Fig. 14 shows the mean and RMS of $E^{\text{rec}}/E^{\text{true}}$ as a function of zenith angle for 10^{19} eV and an array spacing of 1500 m. The results for two hadronic models are given (black markers). Individual results for proton and iron primaries are also shown. Biases for individual primaries are expected since the average N_{μ} and X_{max} for the ensemble are used in the reconstruction. It is interesting to note that the core bias is reduced to be compatible with zero (see Fig. 15; 100 proton showers at $10^{19.5}$ eV and $\theta = 45^\circ$), while a phenomenological reconstruction [13] as applied to data of the Pierre Auger Observatory results in a bias of about 55 m.

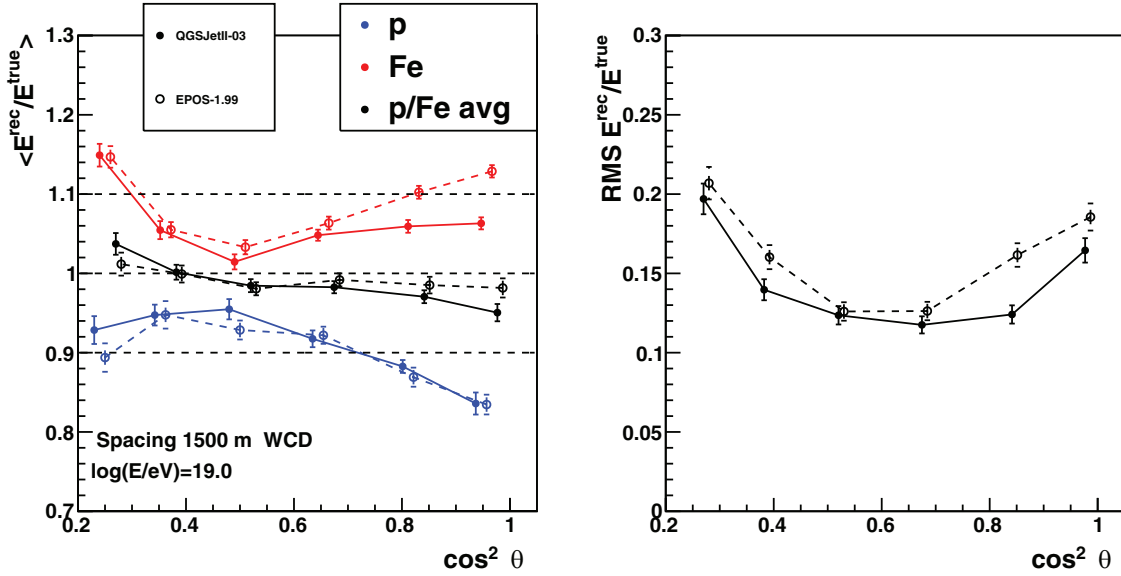


Fig. 14. The mean and RMS of $E^{\text{rec}}/E^{\text{true}}$ as a function of zenith angle for 10^{19} eV. The results correspond to a mixture of 50% proton/iron for two hadronic models. Individual results for proton and iron primaries are also shown.

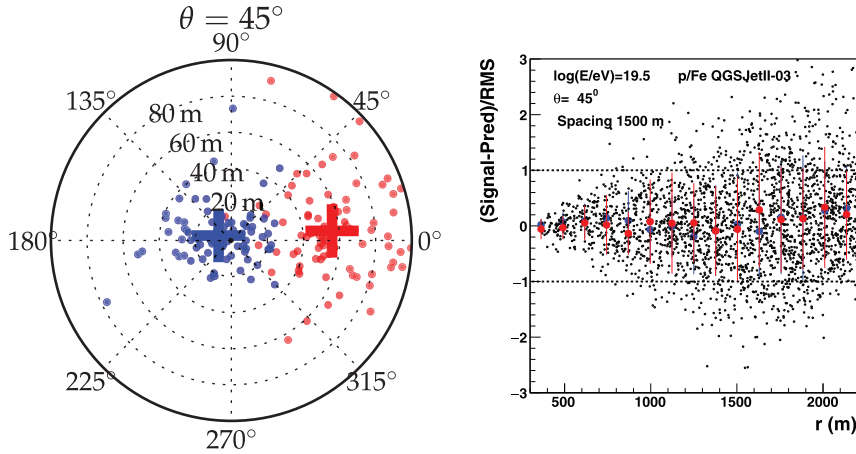


Fig. 15. Left panel: Bias of the reconstructed core position in MC shower plane coordinates for 100 proton simulations at $10^{19.5}$ eV and $\theta = 45^\circ$. Showers are rotated in azimuth such that the incoming direction points to the right. The large cross markers indicate the mean value of all points (red: empirical LDF parameterization [13]; blue: this work). Right panel: the residuals of the signals normalized by the expected signal fluctuations as a function of core distance. Black dots are individual stations in 240 reconstructed proton/iron events. Blue/Red markers is the average for proton/iron primaries (the size of the error bars is the rms). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 16 shows the dependence on energy of the mean and RMS average over all zenith angles. The blue/red shaded areas corresponds to the maximum difference found with the two hadronic models and for two array spacings. The solid and dashed lines correspond to proton and iron showers: a $\pm 10\%$ bias is evident. As expected, the energy resolution for individual primaries is smaller than the ensemble. The energy resolution for iron primaries is smaller than for proton due to the smaller shower to shower fluctuations. The results for iron primaries can be taken as the limit of the method to determine the energy.

The presented procedure to reconstruct the energy of SD events can be validated experimentally in a number of ways:

- Hybrid events: by applying this procedure to hybrid events, the energy can be compared with the FD estimates. A possible energy bias and resolution can be derived.
- The SD rates above the energy threshold at which the SD becomes fully efficient provide an independent way to monitor

changes in the signals with zenith angle, atmospheric conditions, and the age of the array.

- The residuals of the LDF: these should not show any dependency on core distance, zenith angle or energy.

Some of these tests have been applied and validated to/with MC simulations, e.g. see right panel in Fig. 15.

5.3. SD with scintillators on the top of WCD: N_μ^{SD}

The N_μ estimation using hybrid events relies on the FD energy scale. Systematics in this energy scale at the level of 10% yield systematics in $\langle N_\mu \rangle$ at the same level.

In this section we consider an SD array with two types of detectors at each array position: surface scintillators and water-Cherenkov detectors. Since the ratio of the electromagnetic to the total signal is different, it might be possible to estimate N_μ without using the FD energy scale.

Saturation of the time traces either due to the dynamic range of the electronics or detector effects is an issue in this reconstruc-

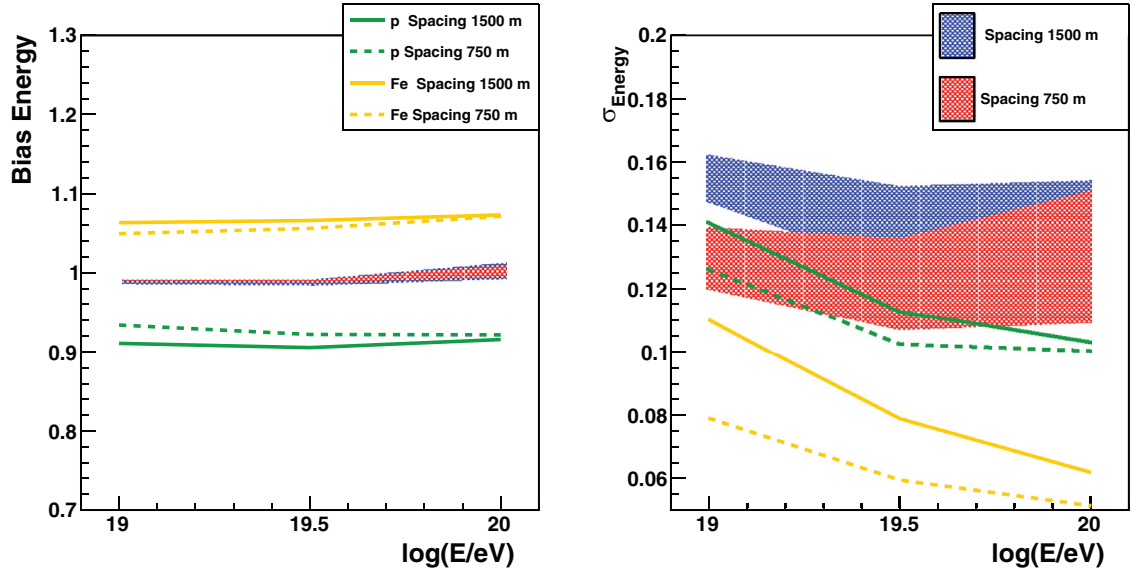


Fig. 16. The mean and RMS of $E_{\text{rec}}/E_{\text{true}}$ averaged over all zenith angles as a function of energy. The results correspond to a 50% mixture of proton and iron showers. The shaded areas corresponds to the maximum difference found with the two hadronic models and for two array spacings. Results for proton/iron separately are also shown.

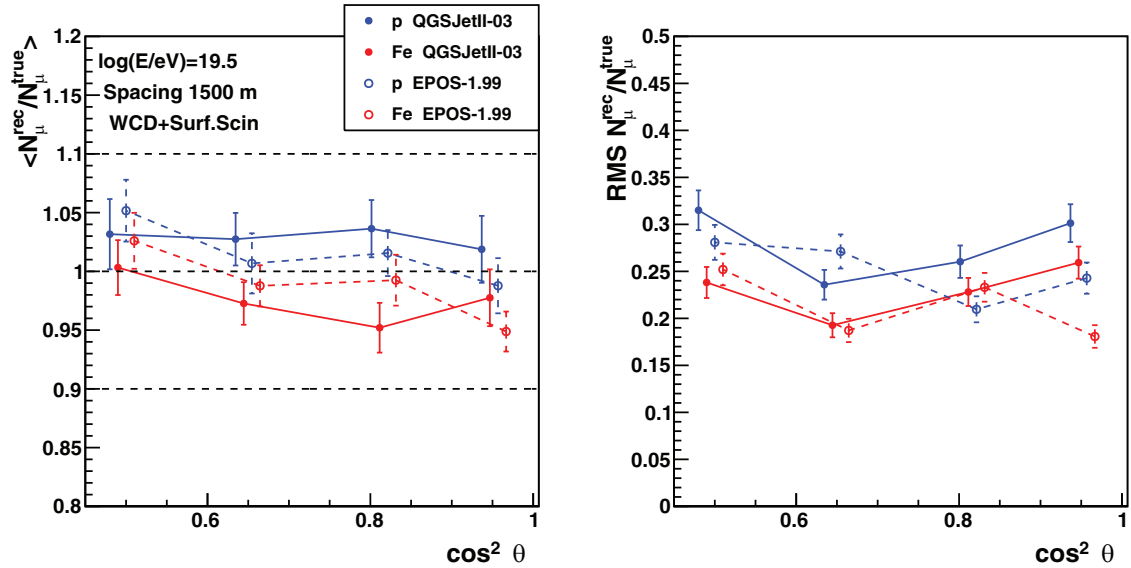


Fig. 17. The mean and RMS of $N_{\mu}^{\text{rec}}/N_{\mu}^{\text{true}}$ as a function of zenith angle at $10^{19.5}$ eV for the SD based N_{μ} reconstruction described in the text. Results for different combinations of primaries and models are shown. The array spacing is 1500 m.

tion. The innermost station is required to settle the energy of the event. We increased the saturation threshold compared to previous results: $\approx 15,000\text{VEM}$ ($30,000\text{VEM}$) for surface scintillators (water-Cherenkov detectors).

The reconstruction procedure is the same as in the energy reconstruction described before, but with energy, N_{μ} , core position and shower direction as free parameters. The X_{max} is left constrained to the true values using a Gaussian distribution with a width of 20 g/cm^2 . We assume that X_{max} is either estimated from FD or from the SD timing [24].

In Fig. 17 the mean and RMS of $N_{\mu}^{\text{rec}}/N_{\mu}^{\text{true}}$ as a function of zenith angle at $10^{19.5}$ eV are shown for different combinations of primaries and models. The array spacing is 1500 m. The resolution of N_{μ} is $\approx 25\%$ and no biases were found. The zenith angle range $0 - 45^\circ$ was explored. Fig. 18 shows the energy dependence of the mean and RMS, averaged over the zenith angles depicted in the previous figure. The uncertainty corresponds to the maximum dif-

ference found with different combinations of primaries and models. Results for two array spacings are shown. The N_{μ} resolution exhibits a strong dependence on array spacing. This is expected: for core distances larger than $\approx 1500\text{m}$, the total signal is mostly due to the muonic component, so N_{μ} and energy become tightly correlated.

This procedure can also be applied to an array of water-Cherenkov detectors and underground scintillators. In this case, the muon detection energy threshold in both detectors is significantly different, which may yield to inconsistencies in the N_{μ} definition if the true energy spectrum of muons differs significantly w.r.t. the reference model and primary.

6. Summary

We introduced a new shower component which accounts for sub-showers of large transverse momentum initiated by neutral pions produced in hadronic interactions at low energy. We explicitly

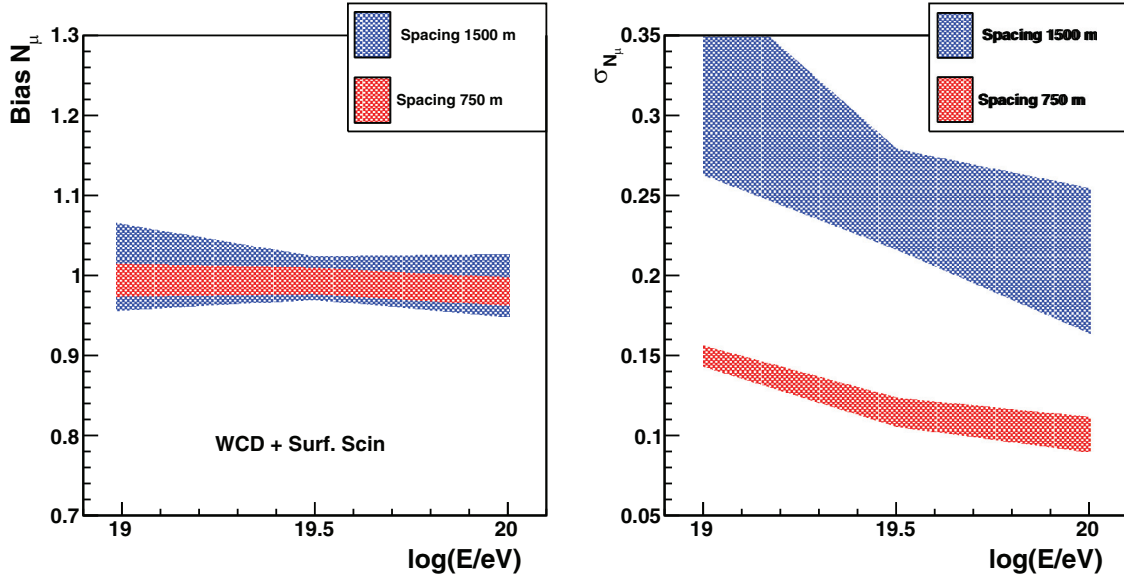


Fig. 18. The mean and RMS of $N_{\mu}^{\text{rec}}/N_{\mu}^{\text{true}}$ averaged over all zenith angles as a function of energy for the N_{μ} SD based reconstruction described in the text. The uncertainty corresponds to the maximum difference found with different combinations of primaries and models. Results for two array spacings are shown.

used the relation between neutral pions (photons) and charged pions (muons) to model this component.

Using this newly introduced component, we developed a parameterization of the signal sizes in Extensive Air Showers for different types of detectors based on proton QGSJetII-03 simulations. The MC simulated signals in a ground detector at (r, Ψ) produced by a shower with a given $(E, X_{\text{max}}, N_{\mu}, \theta, \rho_{\text{ground}}^{\text{air}})$ is described by this parameterization with an accuracy better than 10% in the whole parameter space. Simulations for other primaries and/or hadronic models are described with the same accuracy. This parameterization accounts properly for atmospheric and ground effects and the different atmospheric attenuation of each of the shower components. Shower to shower fluctuations at a fixed energy are accounted for through the per shower value of X_{max} and N_{μ} . Moreover, it is analytical, which allows its usage in minimizations of Likelihood functions during shower reconstruction. The differences between hadronic models and primaries are only due to the different predictions of X_{max} and N_{μ} .

We developed a procedure to derive N_{μ} on an event by event basis using Hybrid events in experiments with SD and FD detectors. Also, and using $\langle N_{\mu} \rangle$ from the previous procedure together with the $\langle X_{\text{max}} \rangle$ measured by FD, we developed a procedure to derive the energy for any SD event regardless if there are coincident FD measurements. The procedures presented are applicable to an array of water-Cherenkov detectors and/or scintillators (surface or underground).

We also developed a procedure to derive N_{μ} on an event by event basis in an SD array (no FD) with two types of detectors at each position: surface scintillators and water-Cherenkov detectors.

Acknowledgement

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isterium für Bildung und Forschung (BMBF) and by the Helmholtz Alliance for Astroparticle Physics.

Appendix A. Cuts based on hadronic generation and weight to separate the jet component

The shower library used in this work was not simulated with the CORSIKA history version [19]. Therefore, another procedure is needed to separate the jet component. The left panel in Fig. A19 shows the distribution of hadronic generation vs. particle weight for electromagnetic particles for the same shower as in Fig. 2. The core distance is 400 m. The z axis is the expected signal in a water-Cherenkov detector. The hadronic generation counter is explained in detail in [19]. The particle weight distribution is affected by the thinning level (10^{-6}) and the maximum weight settings (10000 for electromagnetic particles and 100 for hadrons for $E = 10^{19}$ eV). Particles with a hadronic generation counter larger than 50 are electromagnetic particles from muon decay products or interactions (this counter is artificially increased by 50 when such processes occur). The distribution of the hadronic generation counter smaller than 50 shows two clear peaks: a sharp peak around hadronic generation 12 for particle weights smaller than 500 and an extended peak at larger particle weights.

The sharp peak is due to hadron jets. In this simulation, hadrons have a maximum weight of 100. Electromagnetic particles produced by these hadrons will start propagating with a weight 100. If the grammage through which they propagate is small enough, the weight will not reach the weight limit for the electromagnetic particles.

To test this hypothesis, we used the cuts based on r_{proj} to remove electromagnetic particles from hadron jets. The result is shown in the right panel of Fig. A19. The sharp peak has disappeared, confirming our hypothesis. Once electromagnetic particles from muon interactions and decay are separated, the pure and the jet electromagnetic components will be separated using a diagonal line cut that passes through these two points: (0,5) and $(0.2 \cdot W_{\text{max}}^{\text{EM}}, 11)$.

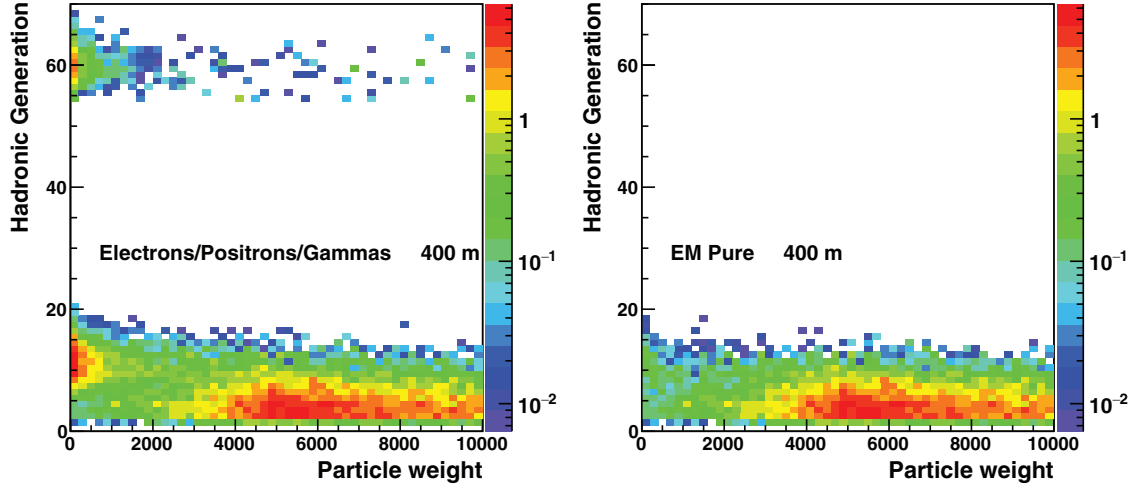


Fig. A19. Left panel: The distribution of hadronic generation counter vs. particle weight for electromagnetic particles for the same shower as in Fig. 2. The sampling area is at a core distance of 400 m. Right panel: same as left panel but removing electromagnetic particles from muon interactions and decay and from hadron jets using the cuts based on r_{proj} .

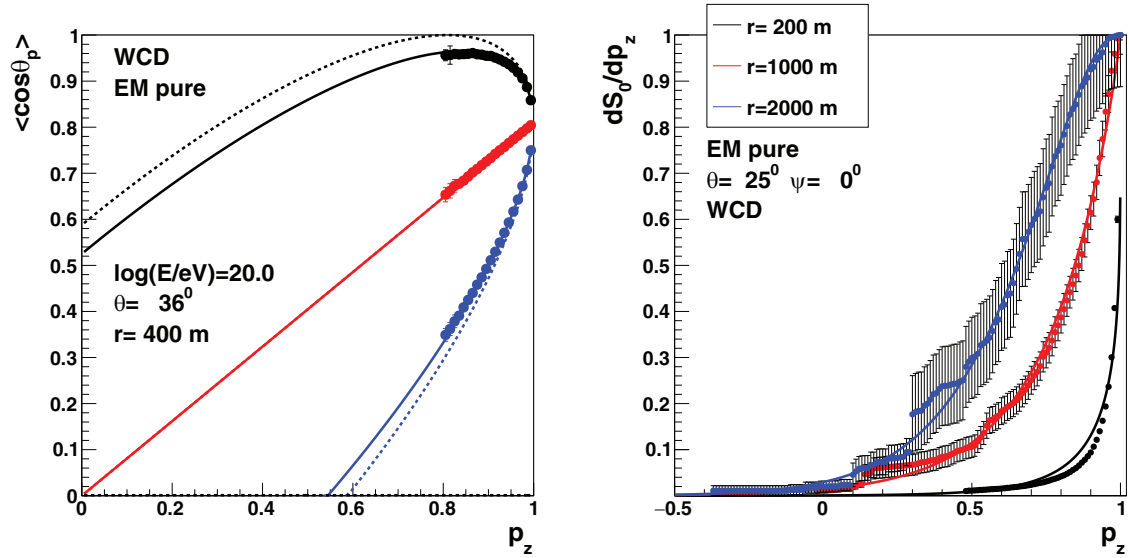


Fig. B20. Left panel: the average $\cos\theta_p$ as a function of p_z for the pure electromagnetic component, see text for more details. Right panel: The cumulative dS_0/dp_z distribution of the pure electromagnetic component for one CORSIKA shower with $\theta = 25^\circ$, see Appendix C for more details.

These cuts are only valid for showers simulated with the thinning level and maximum weights indicated before.

Appendix B. Radial approximation to calculate θ_p

If the shower particles diffuse only radially the zenith angle of a shower particle can be calculated using the following equation:

$$\cos\theta_p = p_z \cos\theta + A_{\text{rad}} \sin\theta \cos\Psi \sqrt{1 - p_z^2}, \quad (\text{B.1})$$

with $A_{\text{rad}} = 1$. The left panel in Fig. B20 shows the average $\cos\theta_p$ as a function of p_z for the pure electromagnetic component from Monte Carlo simulations. The zenith angle is 36° and the core distance 400 m. Different colors correspond to different angles Ψ (0° , 90° and 180°). It was calculated using proton showers at 10^{20} eV (QGSJetII-03) for water-Cherenkov detectors. The dashed lines correspond to the expected value in the radial approximation ($A_{\text{rad}} = 1$). The error bars are the root mean squared. The solid lines are the parameterization adopted in this work by introducing a core distance dependent value of A_{rad} . Similar results are obtained for the scintillation detectors.

Appendix C. Parameterization of $\frac{dS_0}{dp_z}$

$\frac{dS_0}{dp_z}$ will be characterized by a gamma distribution

$$\frac{dS_0}{dp_z}(p_z) = \frac{(1 - p_z)^\alpha e^{-(1-p_z)/\beta}}{\Gamma(\alpha)\beta^\alpha I_0}, \quad (\text{C.1})$$

where I_0 is a normalization factor such that $\int_{-1}^1 \frac{dS_0}{dp_z}(p_z) dp_z = 1$.

The mean and root mean squared of the gamma distribution in the range $(-\infty, 1)$ is given by

$$\langle p_z \rangle = 1 - \alpha\beta \text{ and } \sigma_{p_z} = \sqrt{\alpha\beta}. \quad (\text{C.2})$$

Since p_z is only defined between $(-1, 1)$, $\langle p_z \rangle$ and σ_{p_z} are not physical quantities, but for practical reasons, we use them (rather than α, β) to characterize the shape of the $\frac{dS_0}{dp_z}$ distribution.

In Monte Carlo simulations we only have access to $\frac{dS_0}{dp_z}$ in the range $(p_z^{\text{cut}}, 1)$. However, it is possible to calculate numerically the relationship between $(\langle p_z \rangle, \sigma_{p_z})$ and the mean and root mean squared of the truncated Monte Carlo distribution.

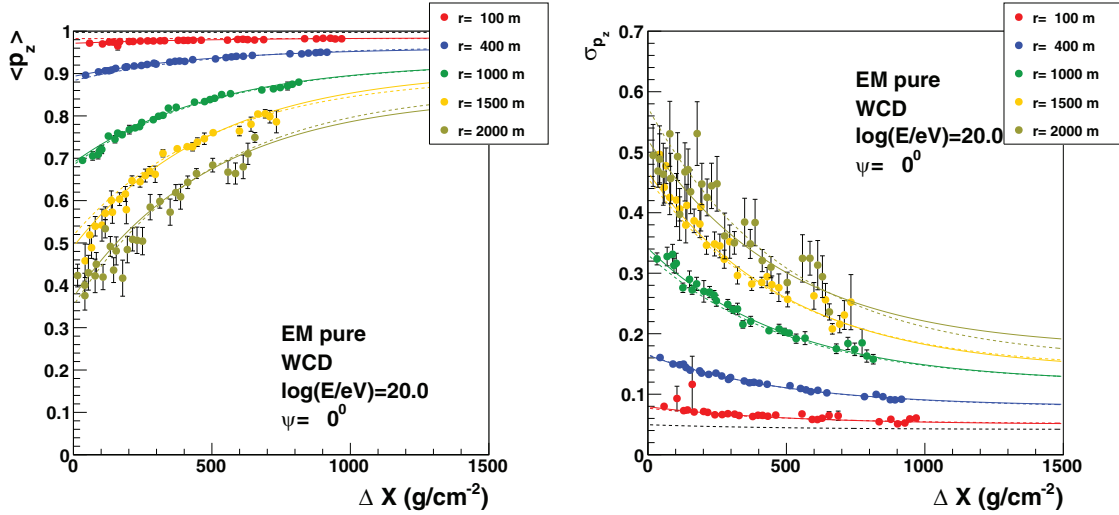


Fig. C21. Left (right) panel: $\langle p_z \rangle$ (σ_{p_z}) as a function of ΔX for the different core distances, the pure electromagnetic component, and the water-Cherenkov detector response.

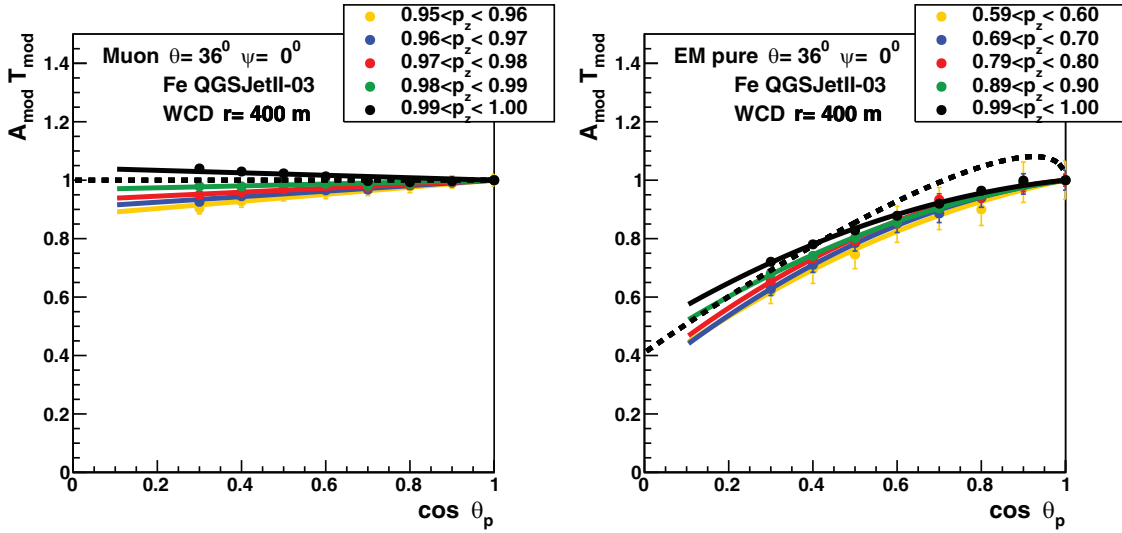


Fig. D22. The dependence of $A_{\text{mod}} T_{\text{mod}}$ on $\cos \theta_p$ using Monte Carlo simulations at $r = 400$ m. The zenith angle is 36° and energy 10^{20} eV. Left (right) panel correspond to the muonic (pure electromagnetic) component. See text for more details.

In the right panel of Fig. B20 the cumulative Monte Carlo distribution for one CORSIKA shower with $\theta = 25^\circ$ is displayed. Sampling areas at different core distances and $\Psi = 0^\circ$ are shown. The pure electromagnetic component for the water-Cherenkov detector is used in this calculation. The solid lines are gamma distributions whose $(\langle p_z \rangle, \sigma_{p_z})$ were calculated using the procedure indicated above and properly normalized to take into account the p_z^{cut} in the Monte Carlo simulations. We further validate the gamma distribution ansatz by comparing the predicted and MC ratio of signals at $\Psi = 0^\circ$ and 180° .

$\langle p_z \rangle$ and σ_{p_z} are then parameterized as a function of $(r, \Delta X)$. Showers with an energy of 10^{20} eV simulated with the reference model (QGSJetII-03) and primary (proton) are used. Only sampling areas with $\Psi = 0^\circ$ and showers with zenith angle larger or equal than 25° are included. In the left and right panels of Fig. C21 examples are shown: $\langle p_z \rangle$ and σ_{p_z} as a function of ΔX for different core distances, the pure electromagnetic component and the water-Cherenkov detector response. The solid lines are the parameterization in ΔX for each core distance. The dashed lines are the parameterizations in $(r, \Delta X)$ adopted in this work. No significant model, primary or energy dependences were found.

Appendix D. Parameterization of $A_{\text{mod}} \cdot T_{\text{mod}}$

$A_{\text{mod}} T_{\text{mod}}(\theta_p | r, p_z)$ is parameterized using MC simulations. Dependences on ΔX were checked and are of second order. 120 CORSIKA showers with a zenith angle of 36° and energy of 10^{20} eV are used. The Ψ angle is 0° , primary iron and the model QGSJetII-03.

In Fig. D22 $A_{\text{mod}} T_{\text{mod}}$ vs. $\cos \theta_p$ is shown for a core distance of 400m using the water-Cherenkov detector response. The left (right) panel corresponds to the muonic (pure electromagnetic) component. Different colors indicate different p_z ranges. In the case of the muonic component, the signal is roughly proportional to the track-length in the detector ($A_{\text{mod}} T_{\text{mod}} = 1$, dashed line). The pure electromagnetic component follows approximately the dashed line that corresponds to $T_{\text{mod}} = 1$ (signal proportional to total energy) with weak p_z dependences.

The dependence of $A_{\text{mod}} T_{\text{mod}}$ on $\cos \theta_p$ is parameterized with a first (second) order polynomial for the muonic (all electromagnetic) components. The free parameters are the values of $A_{\text{mod}} T_{\text{mod}}$ at $\cos \theta_p = 0.3$ (all components), 0.6 (electromagnetic components). These parameters are parameterized as a function of (r, p_z) using simple functions.

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